

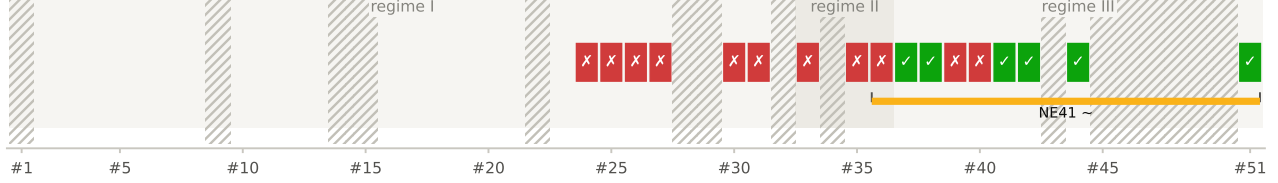
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 2.5/5

Bottom line. Agents respond only from the first model call after a message arrives, and erasing context cuts the coupling, but the nudge kernel is more than the read-out delay: “context is the coupling” fixes where responses land, not their full timing.

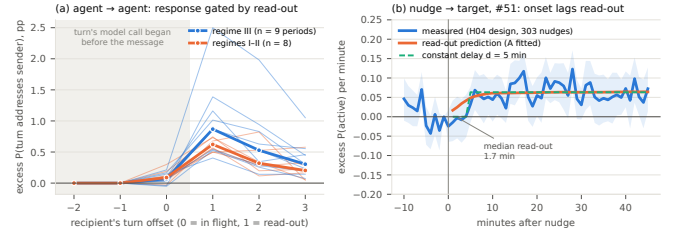
level: hypothesis · pipeline checks 71% · updated 2026-10-04



Goal periods tested (6 supported, 11 failed, 1 mixed). ✓ supported ✗ failed ✓ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

Do agents couple only through what enters their model calls? We tested this in 17 non-holdout goal periods. (1) After a room message, the recipient’s chance of addressing the sender jumps at its first call that started after the message, with no jump at the call already running. This held in 10/11 regime-II/III periods (16/17 post hoc, for recipients not mid-exchange); regime I fails. (2) HH92 fails for nudges: #51 targets read the nudge in 1.7 min (median) but start working at ≈ 5 min, and a constant delay fits as well. (3) Forced context erasure (NE41) cuts addressing of senders read before it by 18% (9/9 periods; predicted 30%). (4) The Claude Code agent’s fetches match a room rule (recall 0.97–1.00) until 2026-03-17, when its feed began replaying April 2025. Hold-out script written, not run.



Observables. (a) Excess chance that a recipient’s turn addresses the sender, by turn offset. Offset 0 is the call already running when the message arrived; offset 1 is the first call after it. Recipients not mid-exchange; thin lines are periods, bold lines regime medians. The response appears only at read-out. (b) The #51 nudge kernel (blue, 95% band) against the zero-parameter read-out prediction (orange) and a 5-min constant delay (green). The response starts about 3 min after read-out.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An agent cannot react until its next model call starts, so the swarm’s coupling delay is its call cadence: tens of seconds when busy, minutes when paused. That follows from how the scaffold is built, so the news is modest; the value is quantitative and diagnostic. Compaction cuts about a fifth of the conversational coupling. A nudge takes about 5 min to show, so don’t expect instant effects. And the replay bug shows exposure can silently come loose from the room, so log what each agent actually fetched. HH92 failed; the gating result rests on mentions.

PHYSICAL MODEL

A two-variable kinetic Ising model (02) with Hawkes kernels (09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room events unread since its last model call. Calls start on the scaffold’s schedule $s_{i,k}$ (after the previous action, or when a pause timer expires). At each call start the field is read and reset, and couplings J_{ij} act only then. W : read-out delay; $h(u)$: post-read-out headroom; o : turn offset; β_F : coupling change after erasure.

$$c_i(t) = \{e : \text{room}(e) = r_i, s_i^- < t_e \leq t\}$$

$$P(a_i | s_{i,k}) \propto \exp \left[a \left(h_i + \sum_{e \in c_i(s_{i,k})} J_{i,\text{src}(e)} \right) \right]$$

$$G(\tau) = A \sum_w p(w) \Theta(\tau - w) h(\tau - w)$$

$$D = G_{\text{addr}}(o=1) - G_{\text{addr}}(o=0)$$

RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

- G24 × failed
 - G25 × failed
 - G26 × failed
 - G27 × failed
 - G30 × failed
 - G31 × failed
 - G33 × failed
 - G35 × failed
 - G36 × failed
 - G37 ✓ supported
 - G38 ✓ supported
 - G39 × failed
 - G40 × failed
 - G41 ✓ supported
 - G42 ✓ supported
 - G44 ✓ supported
- +2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

No round-2 redirects yet.